

Your step-by-step plan to a DevOps job

Simple steps, in the right order. Learn a little every day, build small projects, and you will be job-ready.

LEARN	BUILD	DEPLOY	MONITOR	GROW
Weeks 1 to 8	Weeks 9 to 14	Weeks 15 to 22	Weeks 23 to 25	Weeks 26 to 30

How to use this roadmap

1. Go in order. Every step needs the one before it.
2. For every topic: learn it, then do the task.
3. Put everything you build on GitHub.
4. 1 to 2 hours a day is enough. Just don't stop.

The full map

LEARN	**The basics** Linux · Networking · Git and GitHub · Scripting	Weeks 1 to 8
BUILD	**Package and test your app** Docker · CI/CD · Cloud: AWS	Weeks 9 to 14
DEPLOY	**Run it in the real world** Terraform · Ansible · Kubernetes · GitOps with Argo CD	Weeks 15 to 22
MONITOR	**Watch it and keep it safe** Metrics and dashboards · Logs · Security basics	Weeks 23 to 25
GROW	**Get the job** 4 projects · resume · interviews · certification	Weeks 26 to 30

Times are a suggestion for 1 to 2 hours a day. Faster or slower is fine.

STAGE
LEARN

The basics

Weeks 1 to 8 · Get comfortable with the computer and the internet before any DevOps tool.

01 Linux



2 weeks

What is it? The operating system almost every server runs on. You will use it every day.

LEARN

- [ ] Move around folders: pwd, ls, cd
- [ ] Create, copy, move and delete files
- [ ] File permissions: who can read, write, run (chmod)
- [ ] See running programs and stop them: ps, top, kill
- [ ] Install software: apt install
- [ ] Log in to another computer: ssh

DO THIS

> **Play the free game OverTheWire Bandit, levels 0 to 15.**

FREE RESOURCE

overthewire.org/wargames/bandit

02 Networking

1 week

What is it? How computers talk to each other. Most real problems are network problems.

LEARN

- [ ] IP address and port: the address and the door of an app
- [ ] DNS: turns google.com into an IP address
- [ ] HTTP and HTTPS: how websites send data, status codes like 200 and 404
- [ ] Firewall and load balancer: who can come in, and sharing the traffic
- [ ] Handy commands: ping, curl, dig

DO THIS

> **Write down, step by step, what happens when you open a website.**

FREE RESOURCE

Search YouTube: "how DNS works"

03 Git and GitHub

1 week

What is it? Saves every change in your code, and lets a team work together.

LEARN

- [ ] git init, add, commit, log
- [ ] Branches and merging
- [ ] Push and pull to GitHub
- [ ] Pull Requests: asking a teammate to review your code

DO THIS

> **Put all your practice scripts on GitHub with a short README.**

FREE RESOURCE

git-scm.com/book (free book)

04 Scripting

2 to 3 weeks

What is it? Small programs that do boring work for you. Automation is the job.

LEARN

- [ ] Bash: variables, if/else, loops, running commands
- [ ] Python: basics, files, reading JSON and YAML
- [ ] YAML: almost every DevOps tool uses it

DO THIS

> **Write a script that checks 5 websites and tells you which one is down.**

FREE RESOURCE

learnpython.org

STAGE
BUILD

Package and test your app

Weeks 9 to 14 · Learn how code becomes a ready-to-ship app, automatically.

05 Docker

2 weeks

What is it? Packs your app with everything it needs, so it runs the same everywhere.

LEARN

- [ ] Image vs container: the recipe vs the running dish
- [ ] Write a Dockerfile
- [ ] Run, stop and see logs of containers
- [ ] Docker Compose: run app + database together
- [ ] Push images to Docker Hub

DO THIS

> **Put a small Python or Node app in Docker and run it with a database.**

FREE RESOURCE

docs.docker.com/get-started

06 CI/CD



2 weeks

What is it? Every time you push code, it gets tested and built automatically.

LEARN

- [ ] What CI and CD mean
- [ ] GitHub Actions: a YAML file that runs on every push
- [ ] Jenkins: the older, very popular tool, with a Jenkinsfile
- [ ] Steps: test, build image, push image

DO THIS

> **Make GitHub Actions run tests and build your Docker image on every push.**

FREE RESOURCE

docs.github.com/actions

07 Cloud: AWS

3 weeks

What is it? Renting servers, storage and databases on the internet instead of buying them.

LEARN

- [ ] IAM: users and permissions (do this first)
- [ ] EC2: virtual servers
- [ ] S3: file storage
- [ ] VPC: your private network in the cloud
- [ ] RDS: managed database
- [ ] Load balancer and auto scaling
- [ ] Billing alarm: so you never get a surprise bill

DO THIS

> **Set a billing alarm first. Then run your Docker app on an EC2 server.**

FREE RESOURCE

skillbuilder.aws (free courses)

STAGE

DEPLOY

Run it in the real world

Weeks 15 to 22 · Create servers with code and run apps at scale.

08 Terraform

2 weeks

What is it? Write your servers and network as code, then create them with one command.

LEARN

- [ ] Providers and resources
- [ ] terraform init, plan, apply, destroy
- [ ] Variables and outputs
- [ ] State file: Terraform's memory of what it built

DO THIS

> **Create an EC2 server and an S3 bucket using only Terraform.**

FREE RESOURCE

developer.hashicorp.com/terraform/tutorials

09 Ansible

1 week

What is it? Sets up software on many servers at once, over SSH.

LEARN

- [ ] Inventory: the list of your servers
- [ ] Playbooks: the steps, written in YAML
- [ ] Install packages and copy files to servers

DO THIS

> **Write a playbook that installs Nginx on 2 servers.**

FREE RESOURCE

docs.ansible.com

10 Kubernetes

4 weeks

What is it? Runs many containers, restarts crashed ones and scales up when traffic grows.

LEARN

- [ ] Pod, Deployment, Service
- [ ] kubectl: get, describe, logs, apply
- [ ] ConfigMap and Secret
- [ ] Rolling updates and rollbacks
- [ ] Helm: install a whole app with one command

DO THIS

> **Run your app on a local cluster (minikube or kind) with 3 copies.**

FREE RESOURCE

kubernetes.io/docs/tutorials

11 GitOps with Argo CD

1 week

What is it? Git decides what runs in your cluster. Change Git, and the cluster updates itself.

LEARN

- [ ] Git as the single source of truth
- [ ] Install Argo CD and connect a repo
- [ ] Auto sync and rollback

DO THIS

> **Change the image version in Git and watch Argo CD deploy it.**

FREE RESOURCE

argo-cd.readthedocs.io

STAGE

MONITOR

Watch it and keep it safe

Weeks 23 to 25 · If you can't see it, you can't fix it.

12 Metrics and dashboards



1 to 2 weeks

What is it? Prometheus collects numbers like CPU and errors. Grafana shows them as charts.

LEARN

- [ ] Metrics: CPU, memory, requests, errors
- [ ] Prometheus scrapes your apps every few seconds
- [ ] Grafana dashboards
- [ ] Alerts: get notified when errors go up

DO THIS

Build a dashboard for your app with error rate and CPU.

FREE RESOURCE

prometheus.io/docs

13 Logs

1 week

What is it? Logs tell you why something broke. Collect them all in one place.

LEARN

- [ ] What a log is and log levels (INFO, ERROR)
- [ ] Loki + Grafana, or the ELK stack
- [ ] Search logs for errors

DO THIS

Send your app's logs to Loki and search for ERROR.

FREE RESOURCE

grafana.com/docs/loki

14 Security basics

1 week

What is it? Keep passwords out of code and catch problems before they reach users.

LEARN

- [ ] Never put passwords in code
- [ ] Store secrets in Vault or AWS Secrets Manager
- [ ] Scan Docker images with Trivy
- [ ] Give only the access people need

DO THIS

> **Add a Trivy scan to your CI pipeline.**

FREE RESOURCE

trivy.dev

STAGE
GROW

Get the job

Weeks 26 to 30 · Show what you can do

Build these 4 projects

Projects get you interviews. Give each one its own GitHub repo with a README, a diagram and screenshots.

P1

Website on AWS

A static website on S3, created with Terraform and deployed by GitHub Actions.

Terraform · AWS · CI/CD

- [ ]

P2

App in containers

A frontend, an API and a database running together with Docker Compose.

Docker · Compose

- [ ]

P3

App on Kubernetes

Deploy the app with Helm, let Argo CD update it, and watch it in Grafana.

Kubernetes · Helm · Argo CD · Grafana

- [ ]

P4

Full pipeline

Push code, run tests, build, scan with Trivy, deploy, and get an alert if it fails.

Everything together

- [ ]

Certifications (optional, but they help)

Do them alongside your projects, not instead of them.

1	**AWS Cloud Practitioner** Start here. Cloud basics. No IT background needed.	**$100**
2	**AWS Solutions Architect – Associate** A great next step. Design systems on AWS.	**$150**
3	**AWS DevOps Engineer – Professional** The long-term goal. 2+ years experience recommended.	**$300**

Prices from aws.amazon.com/certification (Nov 2026). Pass one and you get 50% off your next AWS exam. Certifications are valid for 3 years.

Getting the job

RESUME AND PROFILE

- [ ] One page, projects at the top
- [ ] Each project: problem, what you built, tools, GitHub link
- [ ] LinkedIn headline: Aspiring DevOps Engineer | AWS | Docker | Kubernetes
- [ ] Post what you build every week
- [ ] Apply on company career pages and job sites with the fresher filter

PRACTICE THESE QUESTIONS

- [ ] What happens when you open a website?
- [ ] Docker image vs container?
- [ ] What is a Kubernetes Pod and Deployment?
- [ ] Why does Terraform need a state file?
- [ ] How would you build a CI/CD pipeline?
- [ ] A pod keeps crashing. How do you find out why?
- [ ] How do you keep passwords safe?
- [ ] Explain one of your projects from start to end.

Mistakes to avoid

- ✕ Only watching videos and never building anything
- ✕ Jumping to Kubernetes before learning Linux
- ✕ Collecting certificates with no projects
- ✕ Forgetting to stop AWS resources (set a billing alarm!)
- ✕ Paying for "100% job guarantee" courses. Nobody can guarantee a job



You can do this.

Start with Linux today. One topic at a time.
Follow **belikeanu** for simple AWS and DevOps content.